monitoring field conditions, UAVs help maintain uniform crop growth and minimize input costs. Furthermore, yield prediction models enhanced with drone-collected data provide reliable forecasts that support harvest planning, supply chain coordination, and market decision-making. Overall, precision farming enabled by drones enhances both productivity and environmental responsibility.

8.4.2 Remote sensing for soil, water, and crop management

Remote sensing with UAVs has emerged as a vital tool for comprehensive land and crop analysis. Soil characteristics such as texture, moisture content, and organic matter can be evaluated using drones equipped with multispectral and thermal sensors, enabling accurate mapping of soil variability. This assists in determining suitable planting zones, irrigation requirements, and soil treatment strategies.

In irrigation management, UAV thermal imaging identifies areas experiencing water stress, allowing the adjustment of irrigation schedules to prevent over- or under-watering. Similarly, drones enable precise monitoring of crop canopy temperature and root zone moisture conditions, offering real-time data to optimize irrigation efficiency. For pest and disease management, UAV-based vegetation index maps such as NDVI (normalized difference vegetation index) assist in early detection of stress patterns, enabling targeted biological or chemical interventions. Through these capabilities, UAV remote sensing enhances crop health assessment, input optimization, and resource conservation.

Table 8.5 presents a comparative assessment of traditional farming practices and UAV-assisted precision agriculture, emphasizing how drone technology enhances economic efficiency and sustainability in crop production. The table shows that tasks such as field monitoring, irrigation management, and agrochemical application – once reliant on manual labor